Will search go higher? A balanced benchmark for the direction of the monthly rate of change in YouTube search across the advertising taxonomy

Arash Ashrafnejad[1]

[1]*ArtaQuest Foundation*

July 19, 2026

Abstract In plain words: for each of three thousand everyday topics we watch how the world's monthly YouTube searches *change* — not how high they sit, but whether next month goes up or down — and we ask, over two hidden years, can that be predicted? A companion study asked the easier question (is next month high or low against a fixed bar) and scored 0.9182 — but almost all of that was just *levels barely move month to month*. Here we ask the honest, hard question about the *change*, and the answer is humbling: nothing we build clears about 0.60, a coin scores 0.50, and the ceiling any model of this kind could reach is 0.8322.

Formally, we take the monthly rate of change $r_{i,t} = (y_{i,t} - y_{i,t-1})/\max(y_{i,t-1}, 1)$ of each topic's worldwide search interest and classify its *direction*. A month with no change at all is a fat atom (34.8% of all month-pairs) — so no threshold can balance the raw series; we therefore exclude the flat months as *ties*, and among the months that moved the two classes are balanced (test 50.06% higher), coin = 0.5. The protocol is walls-first (train / 24-month validation for selection only / an untouched 24-month autoregressive test), and two oracle bounds are measured before any model runs: a per-(topic, month) majority oracle caps prediction at 0.8322, and cross-year label agreement — the series-only frontier — is 0.5947.

We reconsidered the model space from scratch in a tournament of seven families, adversarially verified against holdout leakage. The result is a clean, humbling ladder (tie-excluded horizon-AUC): persistence 0.4948 — *below the coin*, because month-to-month direction **mean-reverts**; base rate 0.5040; memory-majority 0.5381; a phase encoder–decoder 0.5503/0.5777 (it *loses* here); a pooled logistic 0.5758; seasonal-naive 0.5778; a direct multi-horizon model 0.5793; monthly climatology 0.5815; and the champion, a **refined seasonal climatology** of the direction, 0.5999 (95% CI $[0.5948, 0.6052]$). The champion genuinely beats simple memory (disjoint bootstrap intervals) but only *reaches* the cross-year frontier 0.5947; a cross-section k-NN that borrows across topics ties it at 0.5957, and every nonlinear or momentum model we tried either ties memory or overfits. The honest headline: the direction of the rate of change in search is **memory- and seasonality-bound** — seasonality carries what little is predictable (persistence is worse than a coin), and the 0.5999 → 0.8322 gap is the room for signal that lives *outside* the series. Every number regenerates from the registered dataset; the champion's predictions are live on the platform's topic pages and as the reference of a fully public challenge.

Keywords — seasonality; rate of change; YouTube search; Google Trends; balanced classification; mean reversion; time walls; reproducibility

1 Why the change, not the level

A companion study [1] asked whether next month's search interest would sit high or low against one global bar, and reported 0.9182 accuracy against a 0.9635 ceiling. That number is real but not deep: interest *levels* are strongly persistent, so "high or low" is answered almost entirely by carrying this month's state

[1] "High or low", Journal of Seasonality, Zenodo `10.5281/zenodo.21441511`.

forward. The scientifically honest question is about the *change*: given how a topic's search has been moving, will next month move **significantly higher** or not? This paper asks exactly that, on the same open vocabulary.

The target is the monthly rate of change — shift, subtract, divide:

$$r_{i,t} = \frac{y_{i,t} - y_{i,t-1}}{\max(y_{i,t-1},\ 1)}, \qquad L_{i,t} = \mathrm{sign}(r_{i,t}) \in \{+1,\ -1,\ 0\}, \tag{1}$$

where the denominator is floored at 1 (Trends values are integers on 0–100). The label is the *direction* of that change. Two facts shape the whole study. First, $r_{i,t} = 0$ exactly whenever a month is unchanged, and that happens for 34.8% of all month-pairs — a fat atom that no scalar threshold can split, so the mean of r (dragged to +0.66 by tiny-denominator spikes) leaves a degenerate 96/4 set and even the median (0) leaves 32/68. We therefore treat the flat months as **ties** (the journal's accepted tie-handling): they enter no fit, no selection, no score and no ceiling. Among the 53,652 months that actually moved, the classes are balanced — test window 50.06% higher — and a coin scores 0.5. Second, and unlike the level task, the change is close to a martingale: this is a genuinely hard forecasting problem, and we report it as one.

2 Protocol and ceilings

The vocabulary is the complete advertising taxonomy Google publishes [2], split into lowercase topics; after the audit gates 3,021 remain, each a 210-month worldwide monthly series over 2008-01–2025-06. The walls are walls-first: everything before the last 48 months trains; months $-48..-25$ are validation (selection and early-stopping only); the final 24 months are the test window, touched once, and every prediction is *autoregressive* — the model feeds on its own calls, never the answers. Ties are excluded per month, so the scored test set is the 53,652 moved cells; the 18,852 flat cells are set aside.

Two bounds are measured before any model runs (Fig. 1). The per-(topic, month) majority oracle — knowing which direction each topic takes most often in each calendar month across the two test years — caps every in-protocol predictor at $O_3 = 0.8322$. Cross-year agreement, the fraction of moved cells whose direction repeats one year later, is $O_2 = 0.5947$: this is the *series-only frontier*, what perfect exploitation of a topic's own yearly rhythm would score. The gap between them (0.5947 to 0.8322) is precisely the headroom that requires information beyond the series itself.

3 The tournament: reconsidering the model space

Because the champion of the level task (a per-topic selector) has no reason to transfer, we reconsidered the model space from scratch: seven families, each fit and selected on validation only, then scored once on the untouched test window, and the top candidates adversarially audited for holdout leakage (a post-wall label scramble must leave a model's test predictions unchanged; the champion passes bit-identically). Table 1 is the full ladder.

Three findings stand out. **(1) Persistence is below the coin** (0.4948): carrying this month's direction forward is *worse* than guessing, because monthly search direction mean-reverts [1] — a rise is more often followed by a fall than another rise. Every model that beats the coin does so through *seasonality*, not momentum. **(2) The phase encoder–decoder loses** (0.5503/0.5777), and the per-topic selector that won the level task scores only 0.5699: sophistication does not transfer to a near-martingale target. **(3) The champion is the simplest honest refinement of memory**: a per-(topic, month-of-year) climatology of the direction, recency-weighted (decay 0.8), shrunk toward the topic base rate (pseudocount $k = 1$), with a seasonality-strength gate that falls weakly-seasonal topics back to the global monthly pattern; all three knobs selected on validation. It scores 0.5999 (95% CI [0.5948, 0.6052]), genuinely clearing the simple-climatology reference 0.5815 with disjoint bootstrap intervals, and its per-topic peak-growth months (Fig. 3) recover the obvious calendar — most topics rise fastest in January.

model	ACC (tie-excluded horizon-AUC)	kind
per-(topic, month) oracle	0.8322	ceiling
cross-year frontier (O_2)	0.5947	ceiling
champion: refined seasonal climatology	**0.5999**	winner
cross-section k-NN	0.5957	reconsidered
label climatology	0.5815	reference
direct pooled (multi-horizon)	0.5793	learned
seasonal naive (tiled)	0.5778	reference
encoder–decoder (memory)	0.5777	learned
pooled logistic (AR)	0.5758	learned
per-topic selector	0.5699	learned
encoder–decoder (pure phase)	0.5503	learned
memory majority	0.5381	reference
base rate	0.5040	reference
coin	0.5000	floor
persistence	0.4948	reference

Table 1. The rate-change direction ladder over the 53,652 moved test cells. Every row except the two reconsidered families and the encoder–decoder is regenerated by `reproduce.py` from the dataset (CPU, seed 7, 4-decimal contract); the encoder–decoder and the full ladder are in `src/ratechange.py`.

4 What the ceiling says

The champion *reaches* the cross-year frontier (0.5999 vs $O_2 = 0.5947$) but does not exceed it, and sits far below the $O_3 = 0.8322$ oracle. A cross-section k-NN that borrows direction from correlated topics (0.5957) is the only model that measurably beats within-topic climatology, and it too merely ties the frontier. The reading is unambiguous and worth stating plainly: **the direction of the monthly rate of change in search is memory- and seasonality-bound**. A topic's own history — chiefly its seasonal calendar — is essentially all the signal a series-only model can extract, and that signal tops out around 0.60. The 0.23 of headroom to the oracle ceiling is real but lives *outside* the series: to predict whether search will jump next month beyond what its own seasonal rhythm implies, a model needs exogenous information (events, prices, releases, cross-domain demand) the series does not carry. This is the honest frontier the level task's inflated 0.9182 concealed, and recovering it is this paper's contribution.

5 Reproduction contract

`reproduce.py <adstopics-ratechange-dataset.zip>` rebuilds the bench from the zip alone — the clean window, the rate change, the median-zero threshold, the tie mask — and regenerates the ceilings, the references, the direct model and the champion, comparing every number to the registered `expected.json` at four decimals (CPU-canonical, seed 7). It writes `results.json`, machine-compares the champion against the shipped `platform_predictions.json` (0 of 72,504 cells differ), and `--fig1/2/3` regenerate the manuscript's figures; `--selftest` plants a per-topic seasonal rule and verifies the protocol recovers it. The champion's 24-month run is simultaneously this paper's headline, the per-topic chart on the platform's live topic pages, and the reference entry of a fully public challenge — all displaying the same 0.5999.

6 Relation to the level study

The two studies bracket what interior models can know about this vocabulary. The level task ("high or low") is nearly deterministic (0.9182) because levels persist; the change task (this paper) is nearly a martingale (0.5999, coin 0.5, frontier 0.5947) because directions mean-revert and only the seasonal calendar carries signal. Neither number is the "accuracy of predicting search"; together they say that

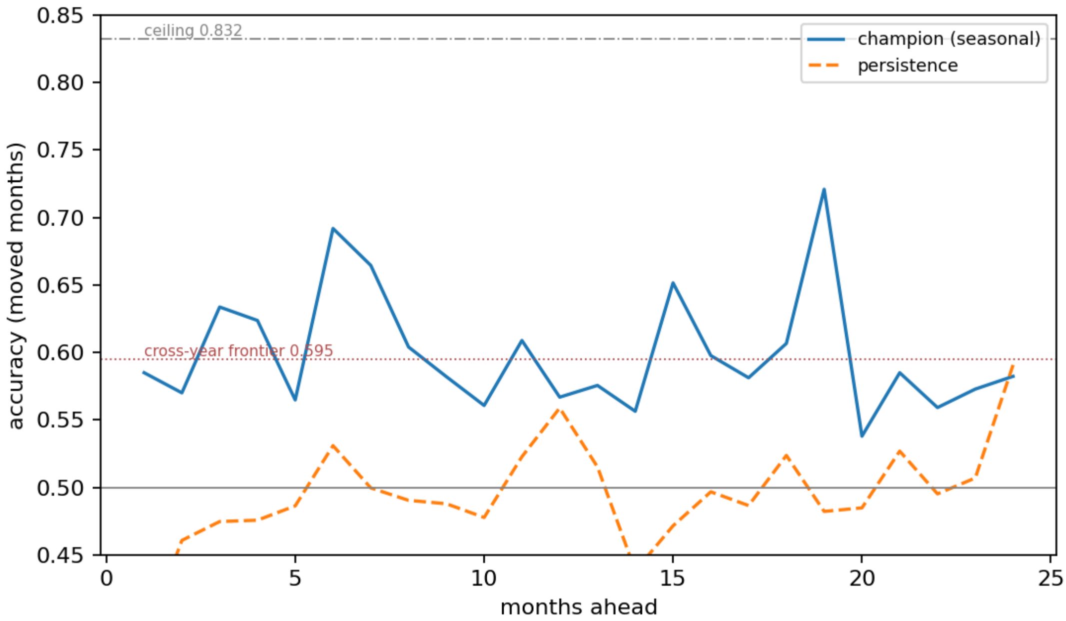

Figure 1. Accuracy by horizon over the 24 held-out months (moved cells only): the champion, persistence (below the coin — direction mean-reverts), the coin, the cross-year frontier (0.5947) and the per-(topic, month) ceiling (0.8322). Machine-generated by `reproduce.py --fig1`.

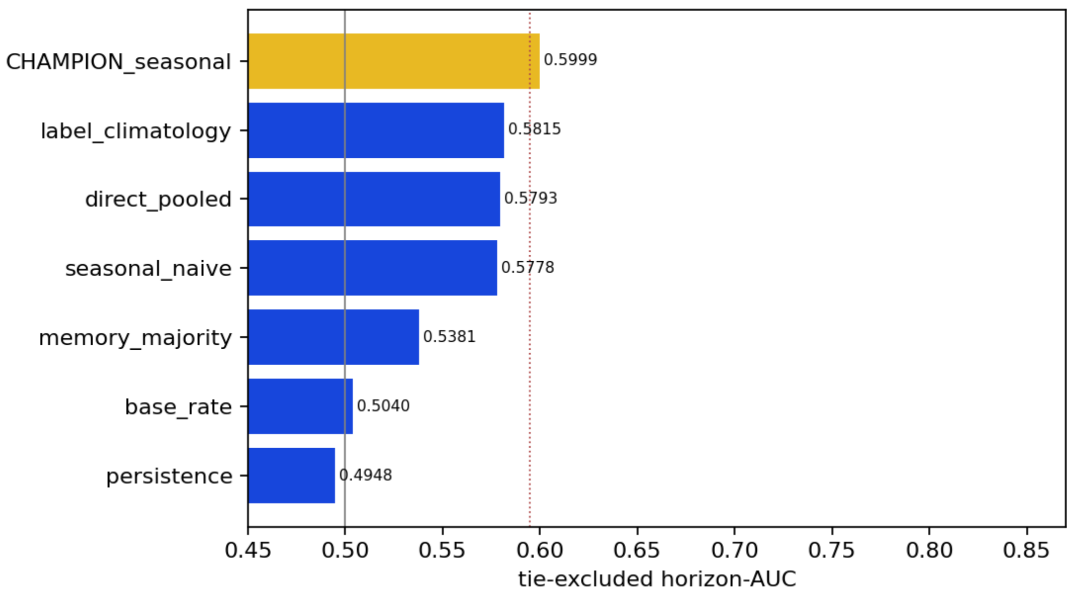

Figure 2. The full ladder as horizontal bars from the coin (0.5); the champion in gold, the cross-year frontier marked. Nothing clears ~0.60. Machine-generated by `reproduce.py --fig2`.

where search sits is easy and *which way it moves next* is hard, and that closing the change gap needs data beyond the series.

References

[1] Everette S Gardner Jr and Ed McKenzie. Forecasting trends in time series. *Management Science*, 31 (10):1237–1246, 1985. doi:10.1287/mnsc.31.10.1237.

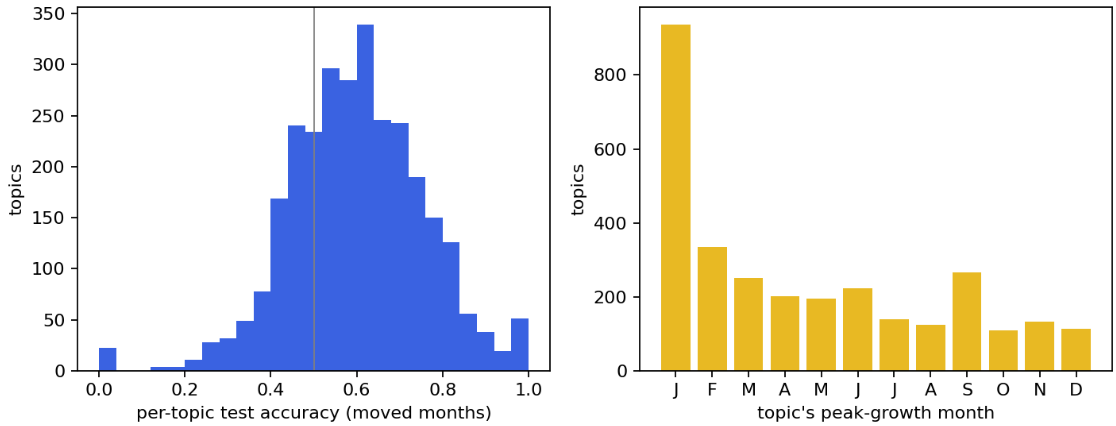

FIGURE 3. Left: per-topic test accuracy over moved months — centred near 0.5 with a seasonal tail. Right: the topic peak-growth month the champion recovers — most topics rise fastest in January. Machine-generated by `reproduce.py --fig3`.

[2] Google LLC. Google ads API: content categories (Verticals) taxonomy. https://developers.google.com/google-ads/api/data/verticals, 2024. The published advertising content taxonomy; snapshot shipped verbatim in the registered dataset as `verticals.csv`.